

Sparse geometry

k NN graph;
shell volumes

Density model

Constrained Gamma
mixture; ICL

Semantic strata

Largest log-rate gap;
explicit background

Core optimization

Exact observed- d_4
MDL event sweep

Background scan

Dyadic windows;
Poisson/Beta code

Extraction

Non-merging watershed;
strict-radius border

Label-free model and threshold selection

Sparse bounded-degree state: $O(nk)$ storage; labels used only by the evaluator